



IIA GROUP ELEMENTS

- Both Be and Al becomes passive on reaction with concentrated nitric acid due to
 - (1) The non – reactive nature of the metal
 - (2) The non – reactive nature of the acid
 - (3) The formation of an inert layer of oxide on the surface of the metals
 - (4) Formation of active layer of oxide on the surface of metals
- In making high strength springs which alloy used ?
 - (1) Cu – Zn
 - (2) Cu – Ni
 - (3) Cu – Be
 - (4) Mg – Zn
- Which of the following cannot be prepared directly from its elements
 - A) BeH_2
 - B) BeF_2
 - C) BeCl_2
 - D) CaH_2
 - (1) All
 - (2) A, B only
 - (3) A, B, C only
 - (4) B, C only
- At temperature below 1200K in vapour state BeCl_2 exist as
 - (1) trimer
 - (2) tetramer
 - (3) dimer
 - (4) Polymer
- Incorrect statement is
 - (1) Stability and solubility of IA and IIA group hydroxides increases down the group
 - (2) Stability and basic strength of IA group hydroxides is more than that of IIA group hydroxides
 - (3) All sulphates of IIA group are insoluble in water
 - (4) Barium nitrate exists as a anhydrous salt
- Statement – I : Setting of plaster of paris involves hydration
Statement – II : Plaster of paris is obtained by heating gypsum to 200°C
 - (1) Both statement I and II are correct
 - (2) Statement I is incorrect and statement - II is correct
 - (3) Statement – I is correct and statement – II is incorrect
 - (4) Both statement – I and II are incorrect
- A metal 'M' readily forms its sulphate MSO_4 which is water soluble. It forms its oxide MO , which becomes inert on heating. It forms its insoluble hydroxide $\text{M}(\text{OH})_2$, which is soluble in NaOH solution. Then the metal 'M' is
 - (1) Mg
 - (2) Ba
 - (3) Ca
 - (4) Be
- Statement – I : Oxides of alkaline earth metals are less basic than oxides of alkali metals
Statement – II : Alkaline earth metals are less electro – positive than alkali metals
 - (1) I and II are true
 - (2) I is true, II is false

- (3) I is false, II is true (4) I and II are false
9. The coordination number of beryllium in $(\text{BeCl}_2)_n$ is
 (1) 4 (2) 2 (3) 6 (4) 3
10. X and Y are metal nitrates. X on heating liberates only O_2 gas but Y on heating liberates NO_2 and O_2 gases. X and Y are respectively
 (1) NaNO_3 and $\text{Mg}(\text{NO}_3)_2$ (2) $\text{Mg}(\text{NO}_3)_2$ and NaNO_3
 (3) NaNO_3 and KNO_3 (4) LiNO_3 and $\text{Mg}(\text{NO}_3)_2$
11. Which of the following is true
 i) Be is not affected by dry air
 ii) Be is amphoteric element
 iii) Be has a maximum covalency four
 (1) i, ii are correct (2) i, ii, iii are correct
 (3) ii, iii are correct (4) i, iii are correct
12. The pair of ions which give the same flame colour in flame test are
 (1) Ca^{2+} and Mg^{2+} (2) Ca^{2+} and Ba^{2+} (3) Li^+ and Sr^{2+} (4) Li^+ and Ca^{2+}
13. Which one of the following statement is correct
 (1) Beryllium exhibits a coordination number of six
 (2) Chlorides of both beryllium and aluminium have chloro bridge dimer structure in vapour phase
 (3) The oxide with more percentage composition in cement is Fe_2O_3
 (4) An adult body contains more mass of Mg when compared to calcium
14. Statement – I : BeSO_4 and MgSO_4 are readily soluble in water
 Statement – II : The greater hydration enthalpy of Be^{2+} and Mg^{2+} ions overcome the lattice enthalpy of BeSO_4 and MgSO_4
 (1) I and II are true (2) I is true, II is false
 (3) I is false, II is true (4) I and II are false
15. Correct statement of the following
 (1) Gypsum decreases the rate of setting of cement
 (2) The order of percentage of ingredients in portland cement is $\text{Ca}_2\text{SiO}_4 > \text{Ca}_3\text{SiO}_5 > \text{Ca}_3\text{Al}_2\text{O}_6$
 (3) Setting of cement is dehydration
 (4) The raw materials for the manufacture of cement are clay and graphite
16. BeF_2 is highly soluble in water whereas the fluorides of other alkaline earth metals are almost insoluble because

- (1) BeF_2 is ionic in nature
- (2) BeF_2 is covalent in nature
- (3) Hydration energy of BeF_2 is much higher than its lattice energy
- (4) Lattice energy of BeF_2 is much higher than that of its hydration energy
17. Which of the following order represents the correct sequence of the increasing basic nature of the given oxides ?
- (1) $\text{MgO} < \text{K}_2\text{O} < \text{Al}_2\text{O}_3 < \text{Na}_2\text{O}$ (2) $\text{Na}_2\text{O} < \text{K}_2\text{O} < \text{MgO} < \text{Al}_2\text{O}_3$
- (3) $\text{K}_2\text{O} < \text{Na}_2\text{O} < \text{Al}_2\text{O}_3 < \text{MgO}$ (4) $\text{Al}_2\text{O}_3 < \text{MgO} < \text{Na}_2\text{O} < \text{K}_2\text{O}$
18. Which of the following statements is false ?
- (1) BeCl_2 is a linear molecule in vapour state above 1200K but it is polymer in the solid state
- (2) Carbides of both Be and Ca react with water to form acetylene
- (3) Oxides of both Be and Al are amphoteric
- (4) Flame test is not given by Mg salts
19. Gypsum is heated to 190°C . The percentage loss in its weight is
- (1) 27% (2) 15.7% (3) 20.9% (4) 36%
20. Correct order among the following
- (1) $\text{BeO} < \text{MgO} < \text{CaO} < \text{BaO} \rightarrow$ Basic nature
- (2) $\text{Be}(\text{OH})_2 < \text{Mg}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Ba}(\text{OH})_2 \rightarrow$ Solubility and basic strength
- (3) $\text{BeH}_2 < \text{MgH}_2 < \text{CaH}_2 < \text{SrH}_2 < \text{BaH}_2 \rightarrow$ Ionic nature
- (4) All
21. The amphoteric hydroxide is
- (1) $\text{Ca}(\text{OH})_2$ (2) $\text{Be}(\text{OH})_2$ (3) $\text{Sr}(\text{OH})_2$ (4) $\text{Mg}(\text{OH})_2$
22. Polymeric covalent hydride and most ionic hydride of IIA group are
- (1) $\text{BaH}_2, \text{BeH}_2$ (2) $\text{BeH}_2, \text{BaH}_2$ (3) $\text{MgH}_2, \text{BeH}_2$ (4) $\text{CaH}_2, \text{BaH}_2$
23. Choose the correct statements from the following
- A) Beryllium is not readily attacked by acids because of the presence of an oxide film on the surface of the metal
- B) Beryllium sulphate is readily soluble in water as the greater hydration enthalpy of Be^{2+} overcomes the lattice enthalpy factor
- C) Beryllium exhibits coordination number more than four
- D) Beryllium oxide is purely acidic in nature
- (1) A and B are correct (2) A and C are correct

- (3) A and D are correct (4) C and B are correct
24. Which of the following should be kept only in the atmosphere of CO_2 to be stable ?
 (1) BeCO_3 (2) MgCO_3 (3) CaCO_3 (4) BaCO_3
25. Chemical A is used for water softening to remove temporary hardness. A reacts with sodium carbonate to generate caustic soda. When CO_2 is bubbled through A, it turns cloudy. What is the chemical formula of A ?
 (1) CaCO_3 (2) CaO (3) $\text{Ca}(\text{OH})_2$ (4) $\text{Ca}(\text{HCO}_3)_2$
26. $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$, $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ lose H_2O molecules on heating. The anhydrous salts become wet when kept in air. These facts indicates that these anhydrous halides
 I) can act as drying agents
 II) As cation size increases the tendency to get hydrated decreases
 III) can absorb moisture from air
 (1) I (2) II, III (3) I, III (4) I, II, III
27. The correct statement among the following is
 (1) Since lattice enthalpy decreases much more than the hydration enthalpy with increase in ionic size the solubility of IIA group hydroxides increases down the group
 (2) Slaked lime is used in white wash due to its disinfectant nature
 (3) The calcium concentration in plasma (100mg/lit) is maintained by calcitonin and parathyroid hormones
 (4) All are correct
28. The hybridization of 'Be' in BeCl_2 in the polymeric solid state, vapour phase and monomeric linear molecule respectively is
 (1) sp , sp^2 , sp^3 (2) sp^3 , sp^2 , sp (3) sp^2 , sp , sp^3 (4) sp^3 , sp^3 , sp
29. Which element contains same number of electrons in the ultimate and penultimate shells
 (1) Be (2) Mg (3) Ca (4) Sr
30. Incorrect statement
 (1) Be and Mg do not exhibit flame colour
 (2) IP_2 of IIA group elements are less than IP_2 of IA
 (3) Hydration energy of M^{2+} ions greater than hydration energy of M^+ ions due to small size and high nuclear charge
 (4) Melting point and boiling point of alkaline earth metals lower than alkali metals
31. Correct statements

- (1) Solubility, stability and basic nature of $M(OH)_2$ increases down the group
- (2) Ionic nature of IIA group metal halides increases down the group
- (3) Solubility of metal carbonates and sulphates decreases down the group
- (4) All are correct
32. Incorrect statement
- (1) BeF_2 is soluble in water
- (2) BeF_2 is prepared by thermal decomposition of $(NH_4)_2 BeF_4$
- (3) Beryllium halides are covalent, hygroscopic and fume in air due to hydrolysis
- (4) $BeCl_2$ is ionic in nature
33. Incorrect trends among the following
- (1) $BeH_2 < MgH_2 < CaH_2 < SrH_2 < BaH_2 \rightarrow$ Ionic nature
- (2) $MgCl_2 > CaCl_2 > BaCl_2 \rightarrow$ Tendency to form halide hydration
- (3) $BeSO_4 > MgSO_4 > CaSO_4 > BaSO_4 \rightarrow$ Solubility order
- (4) $Be(OH)_2 > Mg(OH)_2 > Ca(OH)_2 > Ba(OH)_2 \rightarrow$ Solubility order
34. Which of the following nitrate crystallizes as anhydrous salt
- (1) Ba (2) Mg (3) Ca (4) Sr
35. Correct statements
- (1) For a good quality of cement ratio of silica to Al_2O_3 is 2.5 – 4
- (2) For a good quality of cement ratio of CaO to other oxides is 2
- (3) Major component of Portland cement is tricalcium silicate (Ca_3SiO_5)
- (4) Gypsum is added to cement to increase the process of setting
36. Incorrect statement
- (1) aqueous solution of slaked lime is called lime water
- (2) suspension of slaked lime in water is called milk of lime
- (3) $CaCO_3$ is used as disinfectant
- (4) $CaCO_3$ is used as mild abrasive in tooth paste
37. $Be(OH)_2$ is amphoteric therefore reacts with acid and base to forms 'X' and 'Y' respectively. Then 'X' and 'Y' are
- (1) $[Be(OH)_4]^{2-}$, $[Be(OH)_4]^{2-}$ (2) $[Be(OH)_4]^+$, $[Be(OH)_4]^{2-}$
- (3) $[Be(H_2O)_4]^{2+}$, $[Be(OH)_4]^{2-}$ (4) $[Be(OH)_4]^{2+}$, $[Be(H_2O)_4]^{2-}$

38. Statement – I : A thick suspension of magnesium hydroxide in water is called milk of magnesia which is used as antacid
Statement – II : A clear aqueous solution of slaked lime is called milk of lime
(1) I and II are true (2) I is true, II is false
(3) I is false, II is true (4) I and II are false
39. Barium chloride forms a hydrate $\text{BaCl}_2 \cdot X\text{H}_2\text{O}$. The value of 'X' is
(1) 2 (2) 4 (3) 6 (4) 8
40. The correct statements are
A) Ca, Ba are useful to remove air from vacuum tubes
B) $\text{Mg}(\text{OH})_2$ acts as an antacid
C) Mg + Al alloys are useful in aircraft construction
D) Mg is used for making windows of X – ray tubes
The correct answer is
(1) A, B, C (2) A, B, D (3) B, D (4) A, D
41. Ca^{2+} ions are necessary for
I) Blood clotting
II) Maintaining regular heart beat
III) Muscle contraction
IV) Photosynthesis in plants
(1) I, II, III, IV (2) I, II, III (3) I, II, IV (4) I, III, IV
42. The hormone which regulates concentration of calcium in plasma is
(1) Insulin (2) Calcitonin (3) Parathyroid (4) 2 and 3
43. The best route for the preparation of BeF_2 is
(1) $\text{Be} + \text{F}_2 \xrightarrow{\Delta}$ (2) $\text{BeO} + \text{C} + \text{F}_2 \xrightarrow{\Delta}$
(3) $(\text{NH}_4)_2\text{BeF}_4 \xrightarrow{\Delta}$ (4) All
44. Which of the following does not possess rock salt structure ?
(1) MgO (2) CaO (3) BeO (4) SrO
45. Which property is lower for alkaline earth metals as compared to alkali metals of the same period ?
(1) Hardness (2) Melting point (3) Electropositivity (4) Ionisation potential

CHEMISTRY

1) 3	2) 3	3) 3	4) 3	5) 3	6) 3	7) 4	8) 1	9) 1	10) 1
11) 2	12) 3	13) 1	14) 1	15) 1	16) 3	17) 4	18) 2	19) 3	20) 4
21) 2	22) 2	23) 1	24) 1	25) 3	26) 4	27) 4	28) 2	29) 1	30) 4
31) 4	32) 4	33) 4	34) 1	35) 4	36) 3	37) 3	38) 1	39) 1	40) 1
41) 2	42) 4	43) 3	44) 3	45) 3					

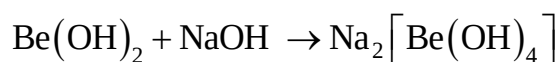
SOLUTIONS

4. dimer in vapour state
polymer in solid state

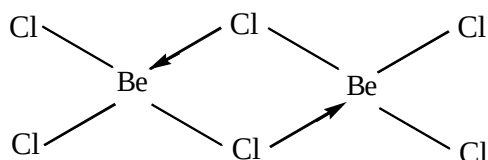
5. BeSO_4 soluble in water

7. $\text{BeSO}_4 \rightarrow$ water soluble

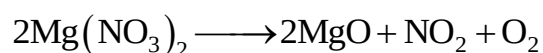
$\text{Be}(\text{OH})_2 \rightarrow$ less soluble



9.

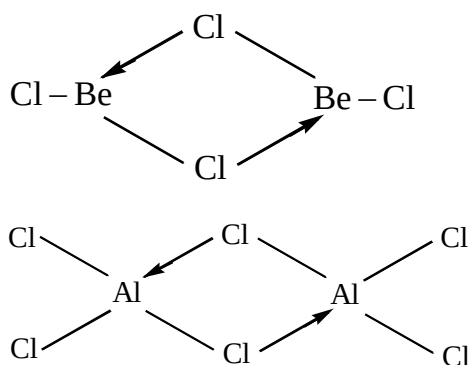


10. $\text{NaNO}_3 \longrightarrow \text{NaNO}_2 + \text{O}_2$



12. Both give crimson red colour

13.



15. Gypsum slows down the speed of setting

16. H.E > L.E (more soluble)

17. Alkali metal oxides are more basic than alkaline earth metals and basic nature increases down the group
18. Be_2C on hydrolysis gives CH_4
19. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \longrightarrow \text{CaSO}_4 + 2\text{H}_2\text{O}$
- 172 gm $\longrightarrow$ 36gm
- 100 $\longrightarrow$?
- $$= \frac{100 \times 36}{172} = 20.9$$
21. Beryllium hydroxide reacts with both acids and bases
22. Ionic nature of hydrides increases down the group
25. $\text{Ca}(\text{OH})_2 \longrightarrow \text{A}$
- $\text{Ca}(\text{OH})_2 + \text{Na}_2\text{CO}_3 \longrightarrow \text{CaCO}_3 + 2\text{NaOH}$
- $\text{Ca}(\text{OH})_2 + \text{CO}_2 \longrightarrow \text{CaCO}_3 + \text{H}_2\text{O}$
29. $1s^2 2s^2$
30. IIA group MP > IA group
32. BeCl_2 is covalent in nature
33. Solubility of hydroxides increases down the group
35. slow down the process of setting
36. $\text{Ca}(\text{OH})_2$ is used as disinfectant
37. $\text{Be}(\text{OH})_2 + 2\text{HCl} + 2\text{H}_2\text{O} \longrightarrow \text{Be}(\text{H}_2\text{O})_4 \text{Cl}_2$
- $\text{Be}(\text{OH})_2 + 2\text{OH}^- \longrightarrow [\text{Be}(\text{OH})_4]^{2-}$
40. Be is used in ray tubes
41. Mg is used in photosynthesis
43. $(\text{NH}_4)_2 \text{BeF}_4 \longrightarrow 2\text{NH}_4\text{F} + \text{BeF}_2$